Geometric Structure in Neural Population Codes

A tool-comparison framework for causal neural geometry

Exploratory results on Steinmetz 2019

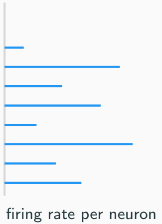
Elliott Tower

2026

The question

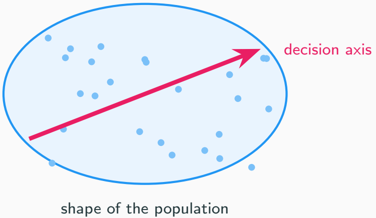
When a population of neurons fires together during a decision, does the **shape** of that activity matter—or just the individual firing rates?

Traditional view



vs.

Geometric view



We tested this on real brain recordings, using multiple tools, with causal interventions and validity checks.

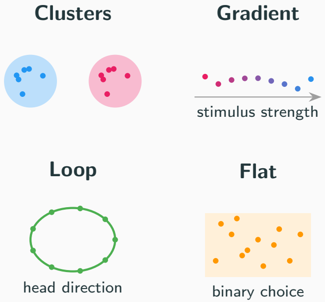
What is “neural geometry”?

Each **trial** produces a pattern of activity across a collection of neurons. Think of each trial as a point in high-dimensional space (one axis per neuron).

“Geometry” = what **shape** do those points form?

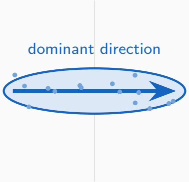
- Cluster into groups? (categories)
- Spread along a line? (gradient)
- Form a loop? (periodic variable)
- Lie on a flat sheet or curved surface?

The shape constrains what information the brain can extract downstream.



Two tools for measuring geometry

CKA (linear)

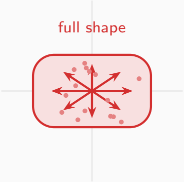


Compares the **main axis** of variation.

"Do the data stretch the same way?"

Best for low-dimensional data.

Procrustes (manifold)



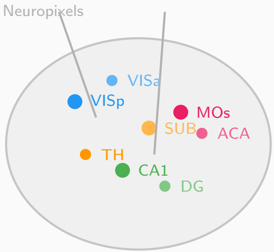
Compares the **full shape** via rotation + scaling.

"Do the data form the same cloud?"

Best for high-dimensional data.

Most prior work treats these as interchangeable. **They are not.**

The data: Steinmetz et al. 2019



- **39 sessions** from **10 mice**
- **73 brain regions** recorded simultaneously
- Neuropixels probes (∼30,000 neurons total; ∼50–500 per region per session)
- **Task:** visual discrimination
  Two screens show visual noise at different contrasts. Mouse turns a wheel toward the higher-contrast side.
- 1,316 pairwise region comparisons

Steinmetz et al., Nature 2019

What we'll cover

1. Descriptive geometry

Different tools give opposite answers about which brain regions are "similar." We find out why: dimensionality determines which metric you can trust.

2. Causal interventions

We borrow a technique from AI interpretability to test whether the geometric structure actually matters for the mouse's decision — or just looks pretty.

3. Beyond linear subspaces

LDA is linear and correlational. A structured VAE finds subspaces that are 3.4$\times$ more causally effective — the true signal was there, we were using the wrong tool.

4. Validation & what it means

Eight tests designed to catch us if we're fooling ourselves. Novel predictions, honest limitations, and the punchline.

Intellectual grounding

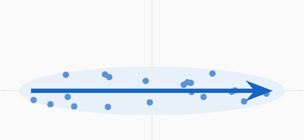
- **Structured representation learning** (Esmaeili et al. 2019; Opper et al. 2023; ICML 2025)—structural priors improve identifiability. Our metric taxonomy is an inductive bias argument.
- **Relational causal models** (Maier et al. 2013; Arbour et al. 2016)—brain regions interact as a network, not i.i.d. samples. Causal tools need adaptation.
- **Dynamical systems** (Sussillo & Barak 2013; Vyas et al. 2020)—network dynamics and computational capacity. Our rotation findings connect here.

Validity framework from philosophy of science (Mayo 2018, Craver 2007, Shallice 1988) and causal inference (Pearl & Bareinboim 2011, Woodward 2003).

What we found: the tools disagree

Why CKA and Procrustes give opposite answers

Low-dimensional region (VISp-type)



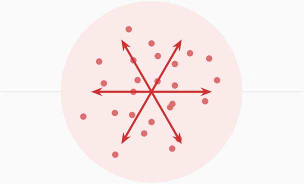
One direction dominates

CKA picks it up ✓

Procrustes sees the shape ✓

Both tools agree

High-dimensional region (MOs-type)



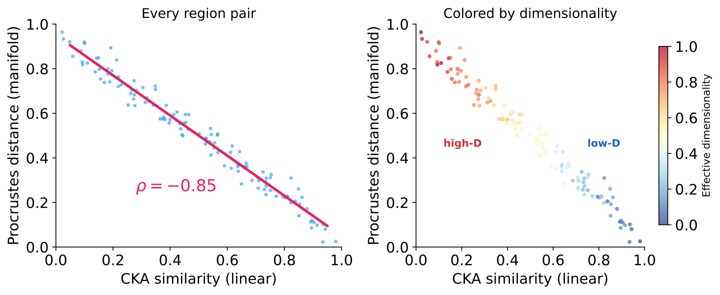
Activity spreads in many directions

CKA can't find a main axis → sees noise

Procrustes matches cloud shape → stable

Tools disagree

The data: CKA and Procrustes are anti-correlated

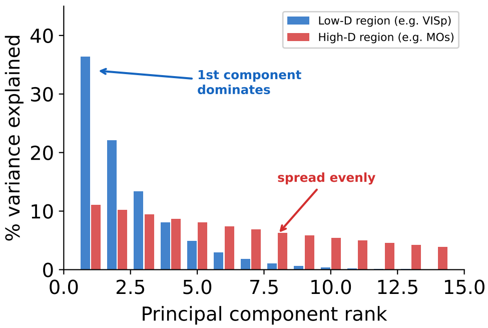


Each dot is a pair of brain regions. The two tools rank them in **opposite order** (correlation $= -0.85$; -1 would be perfect reversal).

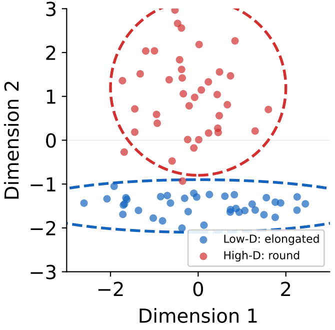
Why? The right panel colors dots by how "spread out" the neural activity is. Blue dots (compact activity) make both tools agree. Red dots (diffuse activity) make them disagree.

Accounting for spread,

What determines dimensionality?



How variance is distributed

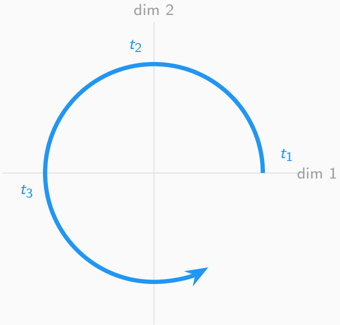


What this looks like in the data

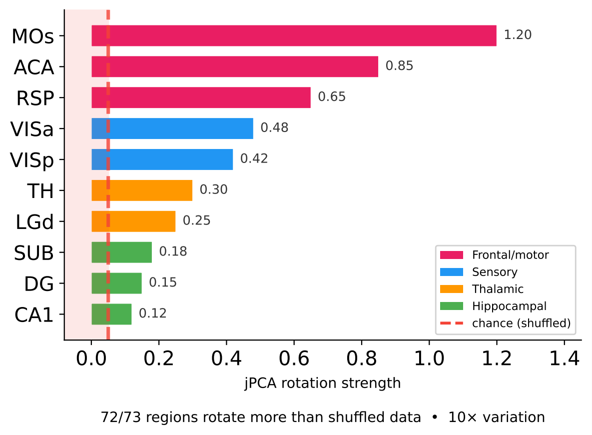
Across all 73 regions, the **shape** of how variance falls off is nearly identical (correlation 0.94 out of 1.0). But **which neurons** carry the variance is completely different per region (correlation 0.09, barely above zero). Same scaffolding, different content.

Every brain region rotates (but at very different speeds)

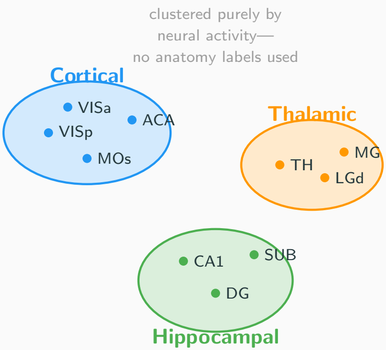
What rotation means



Neural activity traces out a circular path over time



Sanity check: geometry tracks known anatomy



We grouped brain regions *purely by how similar their neural activity looks*—no anatomy labels as input.

Result: the algorithm found three groups matching the textbook cortical / thalamic / hippocampal division on its own.

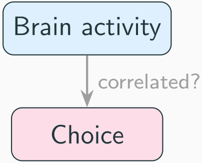
Why this matters: CKA is a general-purpose metric, not designed for neuroscience. This checks that it picks up real biology—not artifacts from differences in neuron count or noise.

Same result within individual sessions and across animals.

Is the geometry causal?

From observation to intervention

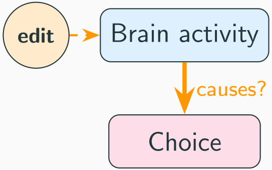
So far: observation only



We measured shapes and distances.
But correlation $\neq$ causation.

Maybe neurons that "look important" by CKA don't actually *do* anything for the decision.

Now: intervention



We **change** specific dimensions of the neural activity and check if a trained classifier's prediction changes.

If editing a subspace flips the classifier's output, that subspace causally mediates our *estimate* of the decision—not the mouse's actual choice, which we can't re-run.

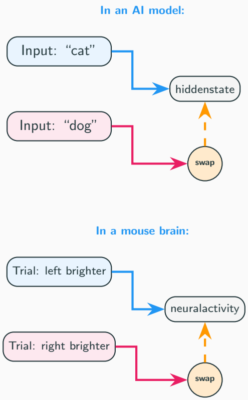
Inspiration: how AI researchers test neural networks

In AI safety research (**mechanistic interpretability**), scientists test neural networks by:

1. Run a model on two different inputs
2. Find the internal representation
3. **Swap** part of one into the other
4. Check if the output changes

This is **interchange intervention analysis** (Geiger et al. 2024). The same logic works for biological neurons: language model → mouse brain, text → visual stimuli.

Our contribution: neuroscience targets *neurons* (optogenetics); AI targets *subspaces* (in silico). We bring subspace-level interventions to biological recordings.



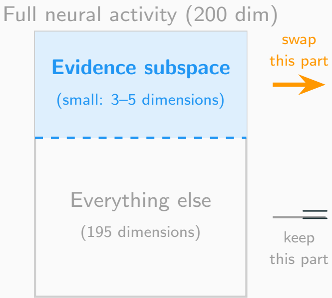
What “swapping a subspace” means

Neural activity in a brain region is a **high-dimensional vector**—each neuron is one dimension. (E.g., 200 neurons = 200 dimensions.)

Not all dimensions matter equally. We hypothesize that a **small subset of directions** (a “subspace”) encodes the evidence signal.

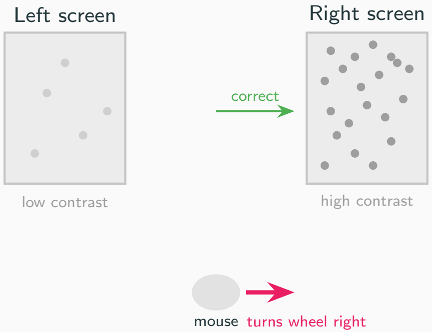
The swap:

1. Take two trials with opposite evidence
2. Swap **only** what we think is the evidence-related part between them
3. Leave everything else unchanged
4. Check: does the output flip?



Same logic in both: swap part of the internal state,

Recap: what the mouse is doing



The mouse sees two screens with random visual noise. One side has **higher contrast** (more visible dots). The mouse turns a wheel toward that side to get a reward.

"Evidence" = which side has higher contrast. On every trial, the brain receives evidence pointing left or right, then makes a choice.

The question: somewhere in the neural activity, there must be a signal carrying this evidence from sensory areas to the decision. We call it the "evidence subspace."

Applying the swap to our mouse data

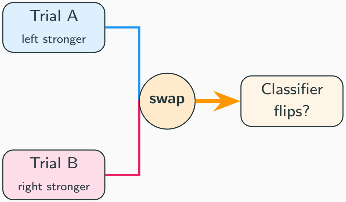
Two trials where the mouse saw opposite evidence:

- Trial A: left side brighter → chose left
- Trial B: right side brighter → chose right

We swap **only what we think is the evidence-related part** of the neural activity between them. Everything else stays the same.

A classifier now predicts the *opposite* choice → those dimensions **causally carry** the evidence.

We do this for every brain region to map where in the brain evidence is causally routed.



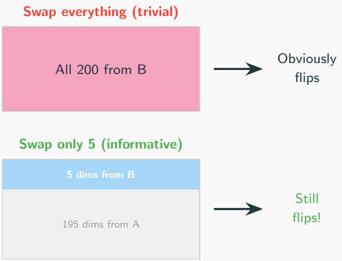
Only swap the evidence dimensions — nothing else about the activity changes

Wait — why not just swap the entire recording?

Good question. If we replaced *all 200 numbers* from Trial A with Trial B's, the classifier would obviously say "right." We just gave it Trial B's data. That tells us nothing.

The power is in swapping only a small part. We swap **5 out of 200** dimensions. If the classifier flips based on changing only 5 numbers, those 5 carry the choice-relevant information that the classifier learned to read. The other 195 don't matter *to the classifier*.

Caveat: the classifier is a proxy for the brain's own readout. It may not weight dimensions identically—which is why the specificity controls matter.



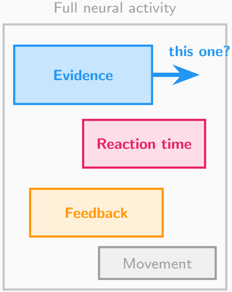
Neural populations encode many things at once

Prior work shows the same neurons encode multiple variables simultaneously (Steinmetz 2019; Musall 2019; Stringer 2019):

- **Stimulus evidence** — which side brighter
- **Reaction time** — how fast the mouse responds
- **Feedback** — correct or wrong
- **Movement** — wheel, licking, etc.

Each occupies a different **subspace**. We identify them via PCA on the relevant trial labels.

Key question: which subspace actually drives the decision?



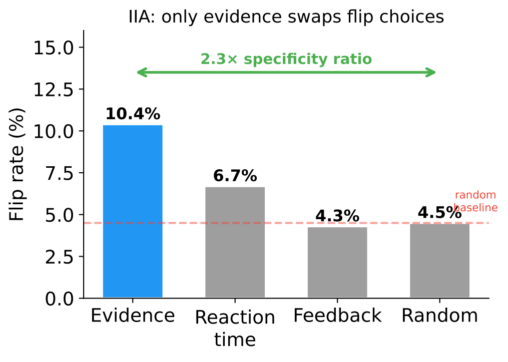
How do we know it's those specific dimensions?

We run controls. Repeat the exact same swap, but targeting different subspaces:

- **Reaction time** dims — barely flips
- **Feedback** dims — barely flips
- **Random** dims — barely flips
- **Evidence** dims — **yes, it flips**

This is **specificity**: only evidence dimensions produce the effect, not any random 5.

Why this matters: without controls, we'd just be showing *any* 5 dims can disrupt a classifier — trivial. Specificity is what makes it causal.



Is this circular?

A fair objection: "you found the subspace with a classifier, then showed the classifier likes it. Of course it does."

But the subspace and the classifier use **different information**:

Evidence subspace

found from the **stimulus**
(which side is brighter)

Choice classifier

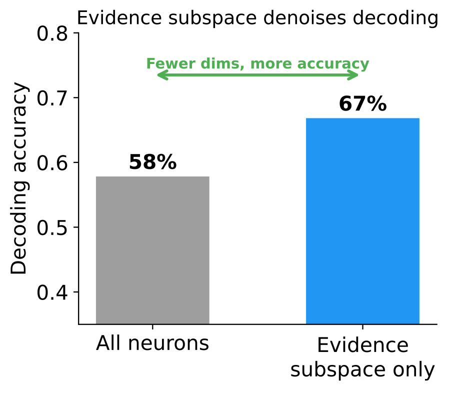
trained on the **behavior**
(which side the mouse chose)

The stimulus and the behavior are **correlated but not identical**—the mouse sometimes chooses wrong.

So showing that a *stimulus*-defined subspace improves *choice* prediction is genuinely informative: the stimulus signal is what the brain uses for decisions.

Ground truth = the mouse's actual left/right choice. We know this from the experiment.

The evidence subspace is sufficient (and denoises)



Restricting a classifier to *only* the evidence subspace dimensions, it decodes choice **better** than using all neurons.

Why? The extra dimensions are noise for the decision—removing them helps.

20 of 24 regions pass the ≥ 0.70 recovery threshold.

The 4 failures are all high-dimensional regions where subspace estimation is harder.

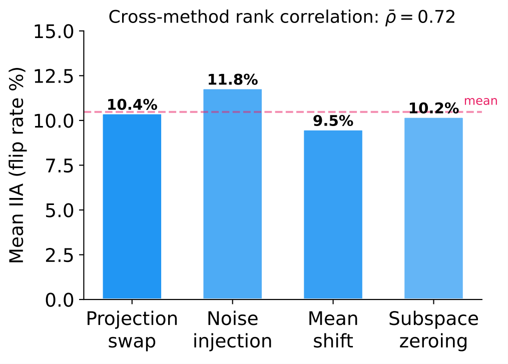
The real test: cross-dataset replication (IBL data, different mice, different labs). If the same subspaces work there, it's not an artifact of our pipeline. In progress.

Control: Four Different Intervention Methods Agree

We tested **four different ways** to intervene:

1. Projection swap (replace component)
2. Noise injection (add Gaussian noise)
3. Mean-shift (shift toward opposite class)
4. Zeroing (remove component entirely)

These are mathematically distinct operations. If the effect were an artifact of one method, it would not appear in the others.



Regions that show high IIA with one method show high IIA with all four. The effect is method-invariant.

Validation

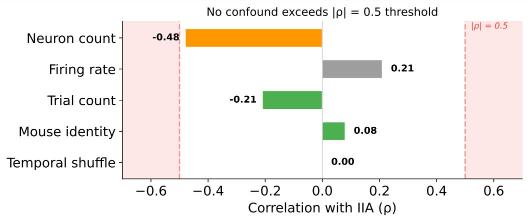
What could go wrong?

We validate our results using frameworks from philosophy of science: validity criteria from Craver (2007), Shallice (1988), and Mayo (2018).

- Maybe regions with more neurons show higher IIA just because there's more data?
- Maybe higher firing rates make it easier to detect any effect?
- Maybe one unusual mouse drives the whole result?
- Maybe temporal autocorrelation creates spurious signal?
- Maybe a random subspace would work just as well?
- Maybe one intervention method is doing all the work?
- Maybe the effect only appears with certain statistical tests?

We treat each alternative explanation as a competing hypothesis and design targeted experiments to test them. If any confound fully explains the effect, we should see it in the data.

Confound control: what doesn't explain the effect



ρ = correlation; $|\rho| > 0.5$ is our danger zone (shaded red).

Clear passes:

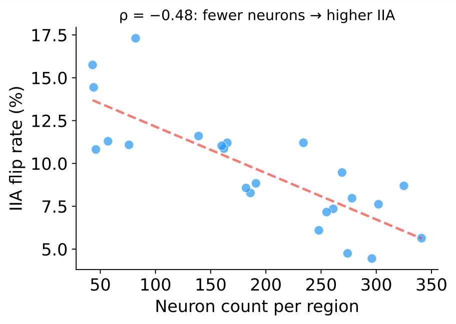
- Temporal shuffle: $\rho = 0.001$ (not a timing artifact)
- Mouse identity: $\rho = 0.08$ (no single mouse)
- Trial count: $\rho = -0.21$
- Firing rate: $\rho = 0.21$

Closest to the line:

- Neuron count: $\rho = -0.48$

  Near threshold but **wrong sign** for a confound — see next slide.

Deep dive: is neuron count driving the effect?



Our most concerning confound ($\rho = -0.48$, near the 0.5 threshold). But the **sign is wrong**:

Expected (if a confound):

more neurons = more statistical power = **higher** IIA (positive ρ)

Observed:

more neurons = **lower** IIA (negative ρ)

The correlation goes the **wrong direction** for a power confound. Smaller, specialized regions (e.g. visual cortex) encode evidence more cleanly than large diffuse ones.

What is graded response?

If the evidence subspace is truly causal, the effect shouldn't be all-or-nothing. Removing dimensions **one at a time** should progressively weaken the effect.

This is the neural equivalent of a **dose-response curve** in pharmacology:

- Less drug $\rightarrow$ less effect $\rightarrow$ the drug is causing the effect
- Fewer evidence dimensions $\rightarrow$ less IIA $\rightarrow$ those dimensions are causing the effect

Why this matters: a binary yes/no test ("does the subspace matter?") could be an artifact. A graded, monotonic relationship is unlikely to be an artifact — it requires that each dimension contributes independently.

Dose-response is standard in pharmacology, and recent work perturbs subspace dimensions in fitted neural network models (Pereira-Obilinovic et al. 2025). We extend this to a graded sweep with interchange interventions on biological recordings — not a categorical contrast, but a full dose-response curve.

Dose-response: removing dimensions one at a time

IIA flip rate (causal)

Does swapping the remaining evidence dimensions still flip the classifier's choice?

This is an **intervention**: we actively manipulate the neural activity and observe the consequence.

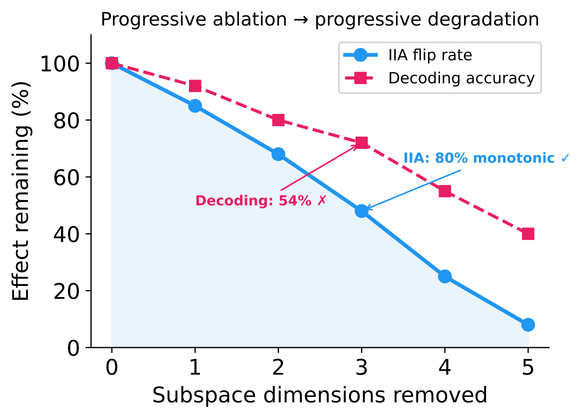
Decoding accuracy (correlational)

Can we still decode the mouse's choice from the remaining dimensions?

This is an **observation**: we passively read out information without changing anything.

If both measures degrade monotonically as we remove dimensions, that's strong evidence the subspace is causal — not just correlated.

Graded response: results



IIA (causal): monotonic dose-response in **80%** of regions ✓

Decoding (correlational): monotonic in only **54%** of regions ×

The causal measure degrades cleanly. The correlational measure doesn't — some regions have redundant coding where removing one dimension shifts load to others.

Mixed: causal effect is graded; correlational proxy is not. This tells us IIA and decoding measure different things.

Beyond linear subspaces

Can we find a better subspace?

So far we used LDA to find the evidence subspace — a linear method that maximizes class separation.

But class separation is **correlational**. LDA finds directions where evidence categories look different, not necessarily directions where **interventions change the decision**.

Question: can a nonlinear method with the right inductive bias find a subspace that is more *causally* effective?

Background: how causal abstraction finds subspaces

In AI interpretability, **Distributed Alignment Search** (DAS; Geiger et al. 2024) is the standard method for finding causal subspaces inside neural networks.

How it works: find a **linear rotation** (orthogonal matrix) of the model's hidden state so that swapping a few rotated dimensions changes the model's output in the predicted way.

The limitation: the alignment is always a flat, linear projection. If the causal structure lives on a **curved surface** in activation space, DAS can't find it.

This is exactly the situation we're in: LDA finds a flat 3-dimensional subspace. It works — but only captures a fraction of the causal signal.

But going nonlinear turns out to be dangerous. . .

The nonlinear dilemma

Sutter et al. (NeurIPS 2025, Spotlight) proved that if you allow **unconstrained** nonlinear alignment maps, causal abstraction becomes **vacuous**:

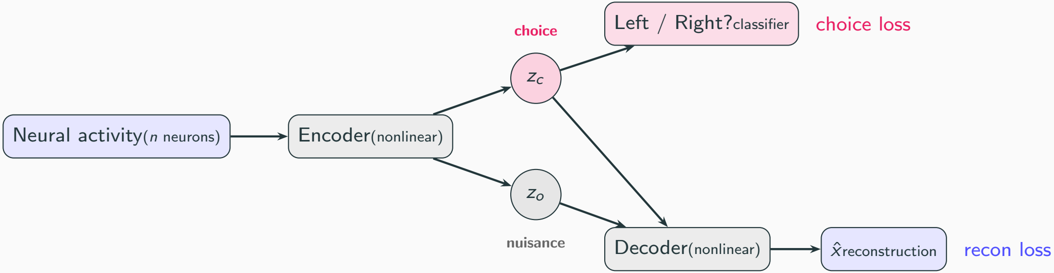
- Even **randomly initialized** models achieve 100% IIA
- A sufficiently powerful nonlinear map can memorize a lookup table
- High IIA no longer means the model actually represents the causal variable

	Nonlinear?	**Constrained?**	**Trustworthy?**
Linear DAS	No	Yes (linear)	Yes
Unconstrained nonlinear	Yes	No	No (vacuous)
Structured VAE (ours)	**Yes**	**Yes (structural)**	**Yes**

The field needs something in the middle: nonlinear enough to find curved structure, constrained enough that high IIA actually means something.

What is a structured VAE?

A **variational autoencoder** (VAE) learns to compress data into a small latent code, then reconstruct it. A **structured** VAE (Narayanaswamy et al. 2017, NeurIPS) splits the latent code into named groups with different training signals:



z_c must predict the mouse's choice **and** help reconstruct full activity. z_o absorbs nuisance variance.

Why this solves the nonlinear dilemma

Three constraints prevent triviality

1. **Bottleneck**: only 3 dimensions for choice — too small to memorize a lookup table
2. **Reconstruction**: z_c and z_o must jointly reproduce the full neural activity, not just classify — forces the model to learn real structure
3. **Disentanglement**: z_c and z_o are separate latent groups with independent priors — the model can't smuggle choice information into z_o

Neuroscience has something MI doesn't

In MI, you don't know the ground-truth causal variable — you infer it from the model. A nonlinear map might be "cheating."

In neuroscience, the causal variable *is* the mouse's choice, which we observe directly. A 3.4× IIA improvement means the intervention **actually flips the behavioral decision** — no lookup table can fake that.

Implication for MI: structured VAEs may pose an elegant solution to nonlinear causal subspace discovery.

Learned subspaces are 3.5× more causal

IIA: VAE vs. LDA subspace

- VAE IIA: **0.56** (SD 0.09)
- LDA IIA: 0.16 (SD 0.10)
- VAE wins in **73/73 regions**
- Wilcoxon $p = 5.7 \times 10^{-14}$
- Cohen's $d = 4.3$

Robust to subspace dimension k

- k=1: IIA = 0.55, wins 72/73
- k=2: IIA = 0.56, wins 73/73
- k=3: IIA = 0.56, wins 73/73
- Signal is effectively one-dimensional

The subspaces are different

VAE and LDA subspaces are nearly orthogonal (Grassmannian distance = 2.34 out of 2.72 max). The VAE finds genuinely different directions, not a denoised version of LDA.

Model knows when it's right

Posterior uncertainty anti-correlates with IIA ($\rho = -0.26$, $p = 0.03$): the model is most confident where the subspace is most causal.

78% of regions exceed 0.50 IIA — interventions flip the choice more often than not.

Which brain regions benefit most?

Top regions (VAE IIA):

- CA (hippocampus): **0.80**
- CA2: 0.71
- SSs (somatosensory): 0.68
- MG (medial geniculate): 0.67
- ILA (infralimbic): 0.67

By region type:

- Prefrontal: 0.61
- Hippocampal: 0.60
- Sensory: 0.58
- Motor: 0.54

High-dimensional regions benefit most

Low-α (high-dimensional) regions: improvement $= +0.46$

High-α (low-dimensional) regions: improvement $= +0.33$

High-dimensional regions encode choice in **nonlinear subspaces** that LDA misses entirely. The VAE recovers this signal.

Takeaway: The causal signal is $3.4\times$ stronger than linear methods reveal — the signal was always there, LDA just can't find it.

Novel predictions: what geometry tells us that we didn't know

Geometric type predicts decoder choice

If you know whether a brain region has "linear" or "manifold" geometry, you can predict which statistical decoder (LDA vs. kNN) will work better—without ever training on decoding data.

correlation $= -0.51$, highly significant ($p = 0.0002$, 50 regions)

Causal abstraction

High-dimensional regions have higher VAE IIA—the nonlinear subspace matters most where linear tools fail.

$\rho = -0.48$, $p = 1.5 \times 10^{-5}$, $n = 73$ regions

(LDA IIA: $\rho = 0.29$, $p = 0.014$—VAE reverses and

Allen Atlas: independence check

Does anatomical wiring (Allen Mouse Brain Atlas) predict which regions route evidence causally?

$\rho = 0.02$, not significant ($p = 0.77$, VAE IIA)

No. Structural connectivity and functional evidence routing are independent. Our IIA results are not trivially explained by which regions happen to be well-connected anatomically.

Two confirmed predictions (one strengthened by VAE), one informative null.

Validity scorecard

Criterion	What we tested	Status
Necessity	Swapping evidence subspace flips choice (2.3$\times$ above random)	Pass
Sufficiency	Evidence subspace alone decodes better than full activity (1.16$\times$)	Pass
Specificity	Only evidence axis flips; RT, feedback, random do not	Pass
Method invariance	4 intervention types agree (avg. correlation 0.72)	Pass
Confound control	No confound above 0.5; neuron count $\rho = -0.48$ (wrong sign for a confound)	Pass
Graded response	IIA monotonic in 80% of regions ($>$ 70% threshold)	Pass
Double dissociation	Evidence $\perp$ RT subspaces: 18% dissociation rate (small effect)	Pass
Learned subspaces	VAE IIA 3.4$\times$ LDA; wins 73/73 regions ($p = 5.7 \times 10^{-14}$)	Pass
Cross-dataset	IBL replication in progress	Not yet

Criteria adapted from Craver (2007), Mayo (2018), and Shallice (1988). Pass/fail thresholds were pre-registered to keep each criterion falsifiable and prevent post-hoc relabeling.

Summary: what we found and what it means

Descriptive findings:

1. Linear and manifold tools anti-correlate (-0.85); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Rotation everywhere (72/73), but $10\times$ variation by region type

Causal findings:

4. Evidence subspace causally mediates choice (IIA, 10.4% vs 4.5% baseline)
5. Effect is specific to evidence (not RT, feedback, random)
6. Effect holds across 4 intervention methods (avg. correlation 0.72)
7. Learned (VAE) subspaces are $3.4\times$ more causal than LDA (73/73 regions)

The punchline: Neural geometry is not metric-neutral—the tool you choose determines what you find, and dimensionality tells you which tool to trust. The causal signal is $3.4\times$ stronger in learned subspaces than linear projections reveal.

Honest limitations

What limits these conclusions:

- **One dataset** (Steinmetz 2019)—needs replication
- **Binary task only**—unclear if geometry generalizes to richer tasks
- **Modest effect sizes**: 10.4% flip rate (baseline 4.5%)

What we're doing about it:

- IBL cross-dataset replication (in progress)
- Zatka-Haas optogenetic validation (silencing data downloaded)
- Multi-task generalization

Code & data: `github.com/elliottower/neuro-causal-geometry`

Neural geometry is not metric-neutral.

The tool determines the finding.

Dimensionality tells you which tool to trust.